

NET512

Designing and Implementing a Secure, Scalable, and Automated Enterprise Network

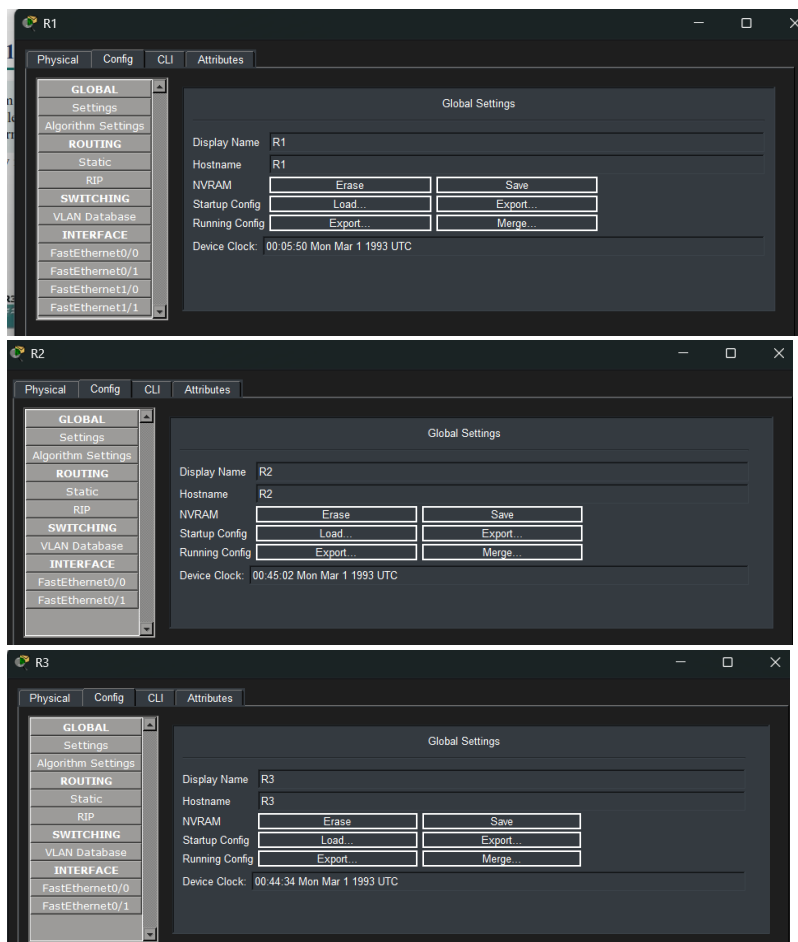
Weekly Practical Plan - NQF Level 5 | 17 Credits

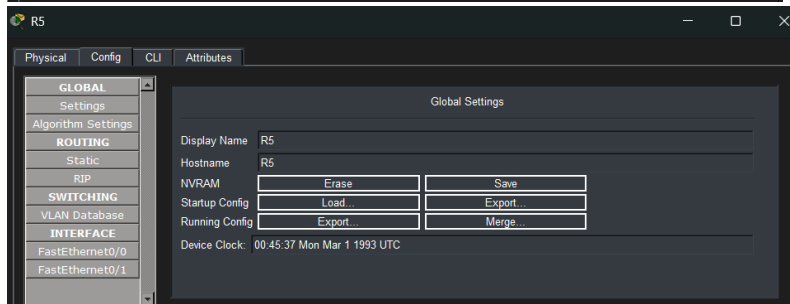
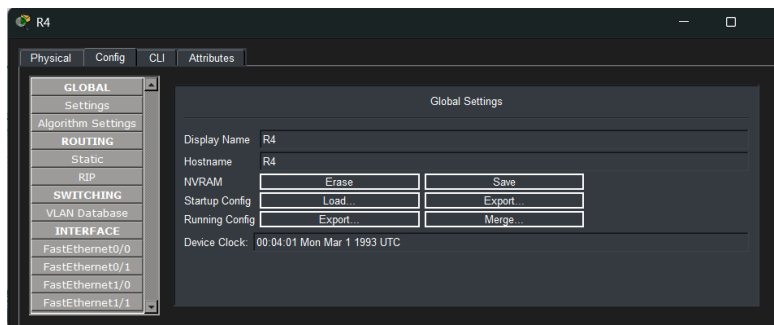
Week 7: Routing Fundamentals (10 Marks)

Tasks:

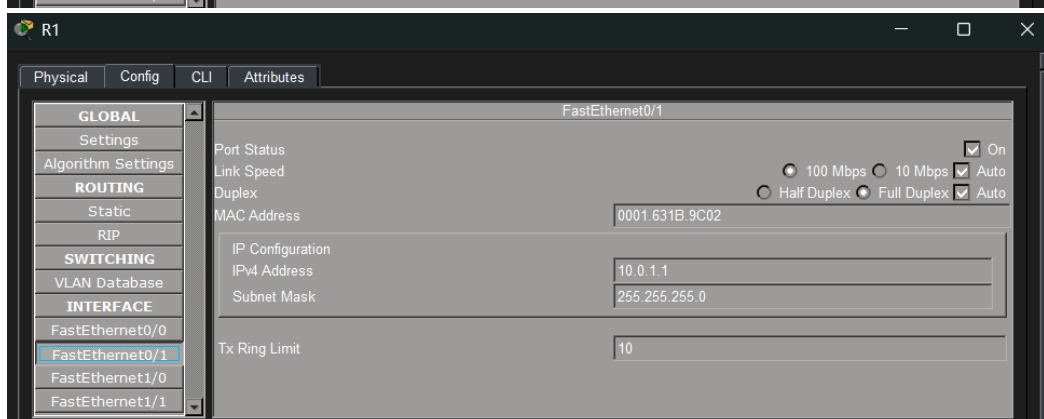
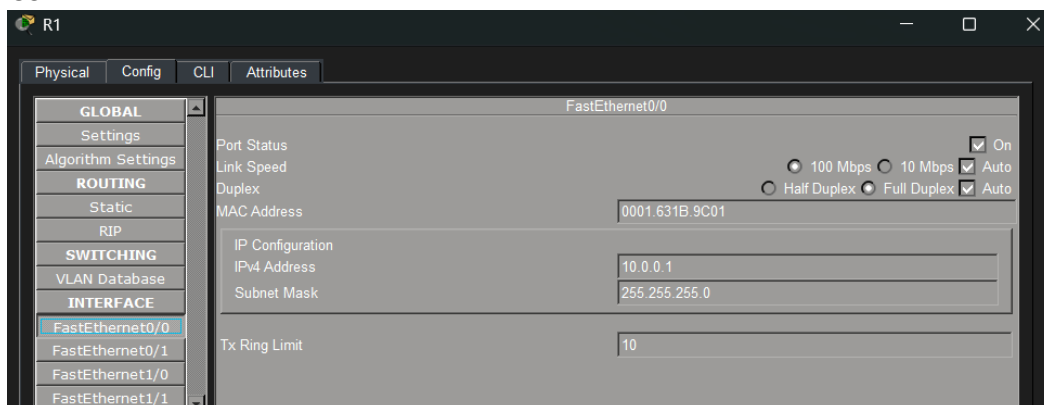
Connected and Local Routes

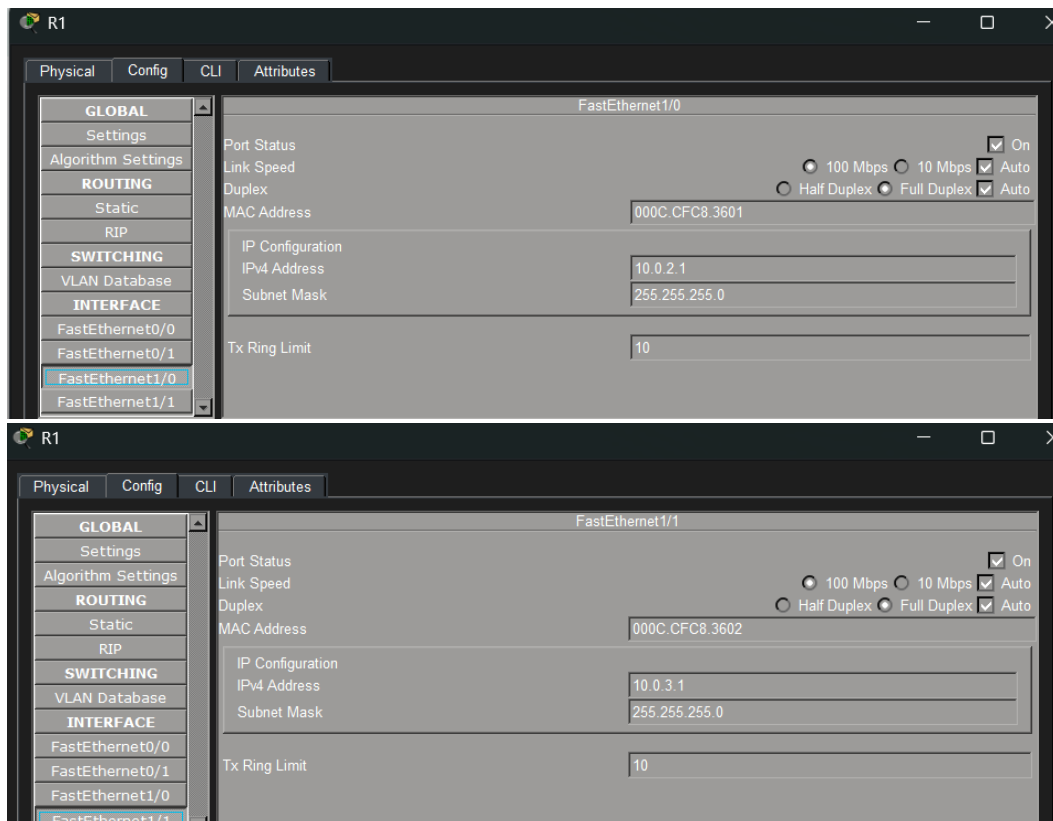
52.





53.





55.

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.0.2.10

Pinging 10.0.2.10 with 32 bytes of data:

Request timed out.
Reply from 10.0.2.10: bytes=32 time<1ms TTL=127
Reply from 10.0.2.10: bytes=32 time=1ms TTL=127
Reply from 10.0.2.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.0.2.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  1  0 ms    1 ms     0 ms     10.0.1.1
  2  0 ms    1 ms     0 ms     10.0.2.10

Trace complete.

C:\>
```

56.

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

57.

```
ip address 10.0.0.2 255.0.0.0
R2(config-if)#ip address 10.0.0.2 255.255.255.0
R2(config-if)#
R2(config-if)#exit
R2(config)#interface FastEthernet0/1
R2(config-if)#no shutdown
R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
ip address 10.1.0.2 255.255.255.0
R2(config-if)#ip address 10.1.0.2 255.255.255.0
R2(config-if)#
R2(config-if)#
R2(config-if)#exit
R2(config)#interface FastEthernet0/0
R2(config-if)#
```

```

R3(config)#interface FastEthernet0/0
R3(config-if)#no shutdown
R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
ip address 10.1.0.1 255.0.0.0
R3(config-if)#ip address 10.1.0.1 255.255.255.0
R3(config-if)#
R3(config-if)#exit
R3(config)#interface FastEthernet0/1
R3(config-if)#no shutdown
R3(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

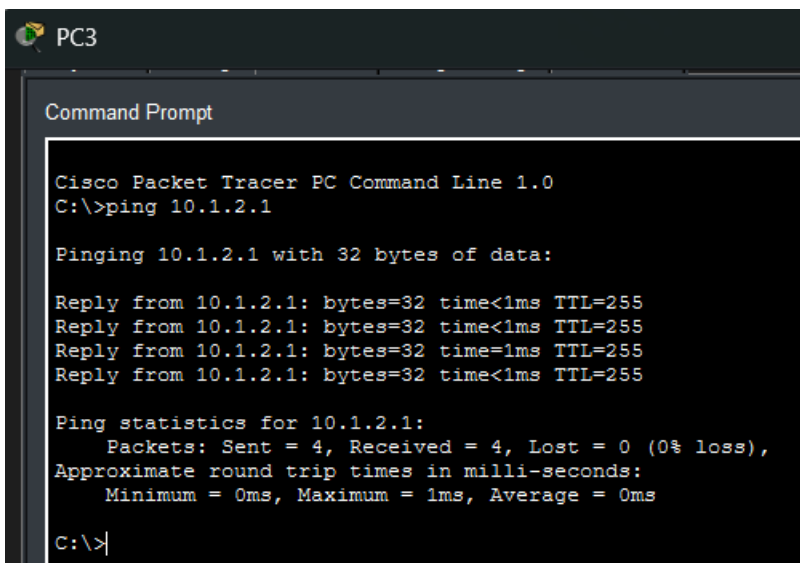
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 10.1.1.2 255.255.255.0
R3(config-if)#ip address 10.1.1.2 255.255.255.0
R3(config-if)#
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#interface FastEthernet0/0
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
ip address 10.1.1.1 255.0.0.0
R4(config-if)#ip address 10.1.1.1 255.255.255.0
R4(config-if)#
R4(config-if)#exit
R4(config)#interface FastEthernet0/1
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
ip address 10.1.2.1 255.255.255.0
R4(config-if)#ip address 10.1.2.1 255.255.255.0
R4(config-if)#
R4(config-if)#exit
R4(config)#interface FastEthernet1/0
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up
ip address 10.1.3.1 255.255.255.0
R4(config-if)#ip address 10.1.3.1 255.255.255.0
R4(config-if)#
R4(config-if)#exit
R4(config)#interface FastEthernet1/1
R4(config-if)#no shutdown
R4(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/1, changed state to up

```

58.



```

PC3
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.2.1

Pinging 10.1.2.1 with 32 bytes of data:

Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255
Reply from 10.1.2.1: bytes=32 time=1ms TTL=255
Reply from 10.1.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.1.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

59.

```
R1(config)#ip route 10.1.0.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.1.0 255.255.255.0 10.0.0.2
R1(config)#ip route 10.1.2.0 255.255.255.0 10.0.0.2
R1(config)#
Enter configuration commands, one per line. End with CNF
R2(config)#ip route 10.0.1.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.0.2.0 255.255.255.0 10.0.0.1
R2(config)#ip route 10.1.1.0 255.255.255.0 10.1.0.1
R2(config)#ip route 10.1.2.0 255.255.255.0 10.1.0.1
R2(config)#
Enter configuration commands, one per line. End with CNF
R3(config)#ip route 10.0.0.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.1.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.0.2.0 255.255.255.0 10.1.0.2
R3(config)#ip route 10.1.2.0 255.255.255.0 10.1.1.1
R4(config)#ip route 10.0.0.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.1.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.0.2.0 255.255.255.0 10.1.1.2
R4(config)#ip route 10.1.0.0 255.255.255.0 10.1.1.2
R4(config)#
```

60.

```
C:\>tracert 10.1.2.10

Tracing route to 10.1.2.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.1.1
  1  0 ms    0 ms    0 ms    10.0.0.2
  2  0 ms    0 ms    0 ms    10.1.0.1
  3  0 ms    0 ms    0 ms    10.1.1.1
  4 10 ms    0 ms    0 ms    10.1.2.10

Trace complete.

C:\>tracert 10.0.2.10

Tracing route to 10.0.2.10 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.1.1
  1  0 ms    0 ms    0 ms    10.0.2.10

Trace complete.

C:\>
```

61.

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.
Reply from 10.0.1.1: Destination host unreachable.

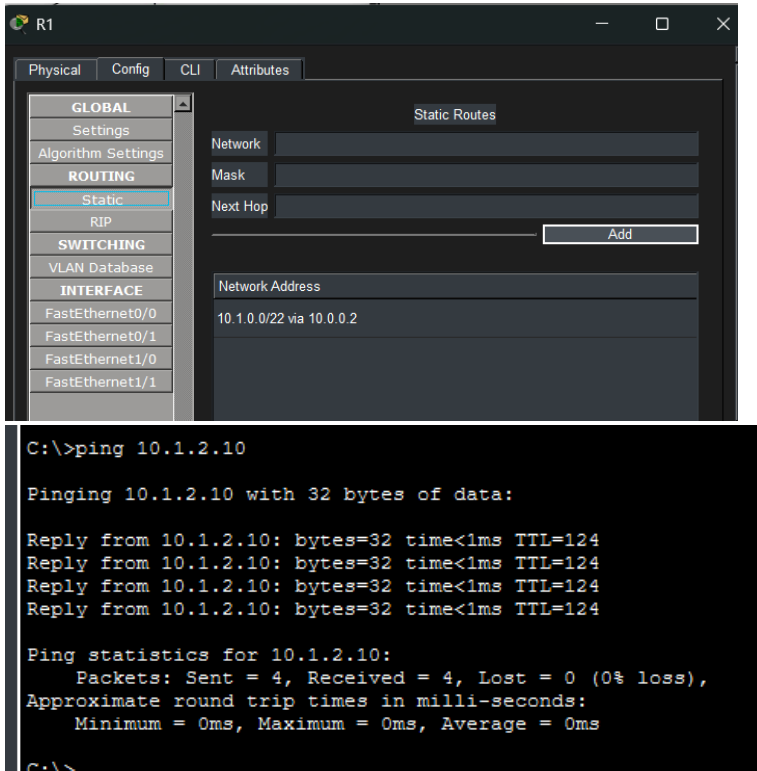
Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

62.

```
R1(config)#ip route 10.1.0.0 255.255.252.0 10.0.0.2
```

63.



The screenshot shows a network configuration window for R1. The 'Static Routes' tab is active, displaying a table with one entry: Network Address 10.1.0.0/22 via 10.0.0.2. Below the table is an 'Add' button. To the left is a sidebar with navigation options: GLOBAL, Settings, Algorithm Settings, ROUTING, Static (selected), RIP, SWITCHING, VLAN Database, INTERFACE, FastEthernet0/0, FastEthernet0/1, FastEthernet1/0, and FastEthernet1/1.

```
C:\>ping 10.1.2.10

Pinging 10.1.2.10 with 32 bytes of data:

Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124
Reply from 10.1.2.10: bytes=32 time<1ms TTL=124

Ping statistics for 10.1.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

64.

Device Name: R5			
Custom Device Model: 2811 IOS15			
Hostname: R5			
Port	Link	VLAN	IP Address
FastEthernet0/0	Up	--	10.1.3.2/24
FastEthernet0/1	Up	--	10.0.3.2/24

65.

```
C:\>tracert 10.1.3.2

Tracing route to 10.1.3.2 over a maximum of 30 hops:

  1  0 ms      0 ms      0 ms      10.0.1.1
  2  0 ms      0 ms      4 ms      10.0.0.2
  3  0 ms      *          1 ms      10.0.0.2
  4  *          0 ms      *          Request timed out.
  5  0 ms      *          0 ms      10.0.0.2
  6  *          0 ms      *          Request timed out.
  7  1 ms      *          0 ms      10.0.0.2
  8  *          1 ms      *          Request timed out.
  9  0 ms      *          0 ms      10.0.0.2
 10 *          0 ms      *          Request timed out.
 11 1 ms      *          1 ms      10.0.0.2
 12 *          0 ms      *          Request timed out.
 13 0 ms      *          1 ms      10.0.0.2
 14 *          0 ms      *          Request timed out.
 15 0 ms      *          0 ms      10.0.0.2
 16 *          0 ms      *          Request timed out.
 17 0 ms
```

66.

```
R1(config)#ip route 10.1.3.0 255.255.255.0 10.0.3.2
R1(config)#
```

67.

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Enter configuration commands, one per line. End with Ctrl-D.
R5(config)#ip route 10.0.1.0 255.255.255.0 10.0.3.1
R5(config)#

```
C:\>ping 10.1.3.2

Pinging 10.1.3.2 with 32 bytes of data:

Reply from 10.1.3.2: bytes=32 time<1ms TTL=254
Reply from 10.1.3.2: bytes=32 time<1ms TTL=254
Reply from 10.1.3.2: bytes=32 time<1ms TTL=254
Reply from 10.1.3.2: bytes=32 time<1ms TTL=254

Ping statistics for 10.1.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

```
C:\>tracert 10.1.3.2

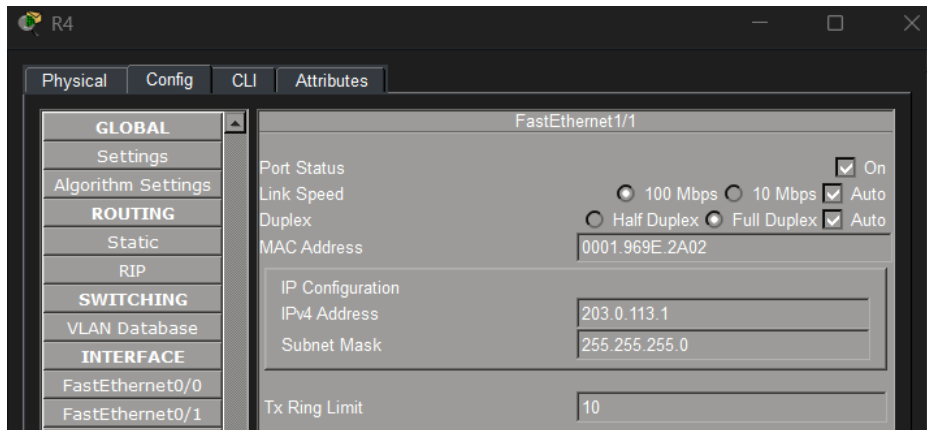
Tracing route to 10.1.3.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.0.1.1
  1  0 ms    0 ms    0 ms    10.1.3.2

Trace complete.

C:\>
```

68.



69.

```
R1(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.1
R2(config)#ip route 0.0.0.0 0.0.0.0 10.1.0.1
R3(config)#ip route 0.0.0.0 0.0.0.0 10.1.1.1
R5(config)#ip route 0.0.0.0 0.0.0.0 10.1.3.1
```

70.

```
R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.3.2
```